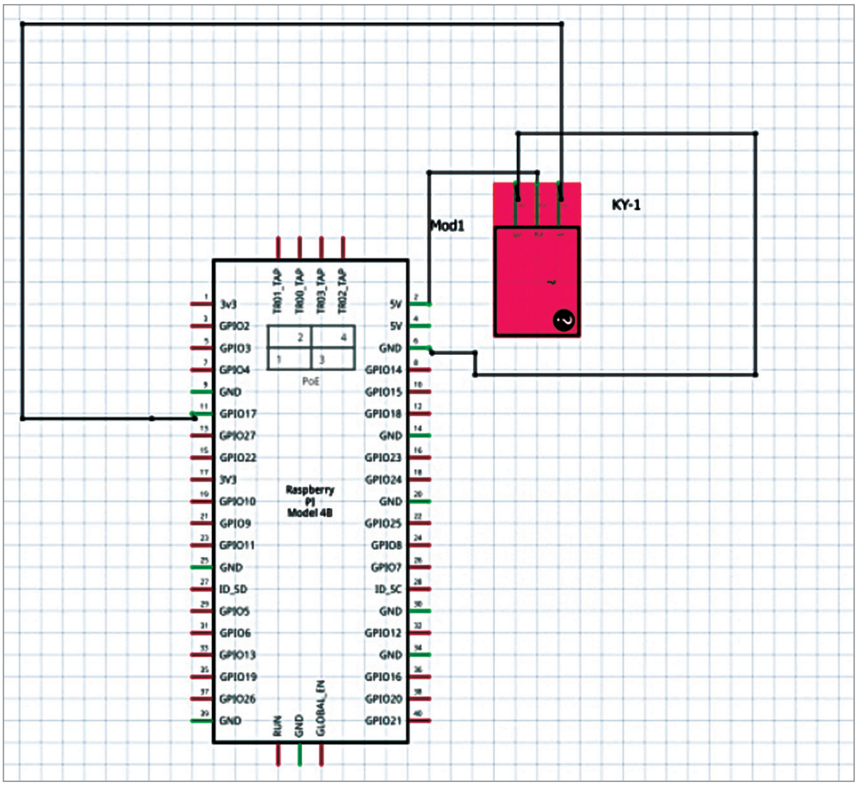

Fig. 6: Connection with relay and Rpi

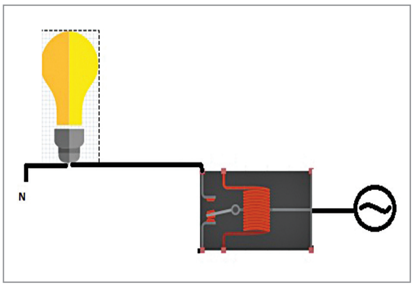

Fig. 7: Connection with relay and bulb

some problems with Python 3, Python 2 has been used instead with the help of command:

```
sudo pip install NeroPy (for python 2)
```

Because the EEG sensor is already paired with the Raspberry Pi, you can read the connected EEG sensor data. Now wear the EEG sensor on your head. To test the EEG sensor data, clone the library and then run the test example code using following sudo git clone:

```
https://github.com/lihas/NeuroPy
cd NeuroPY/NeuroPy
Python test.py
```

The library will help you obtain the following EEG sensor data:

Low theta waves
High theta waves
Low beta waves

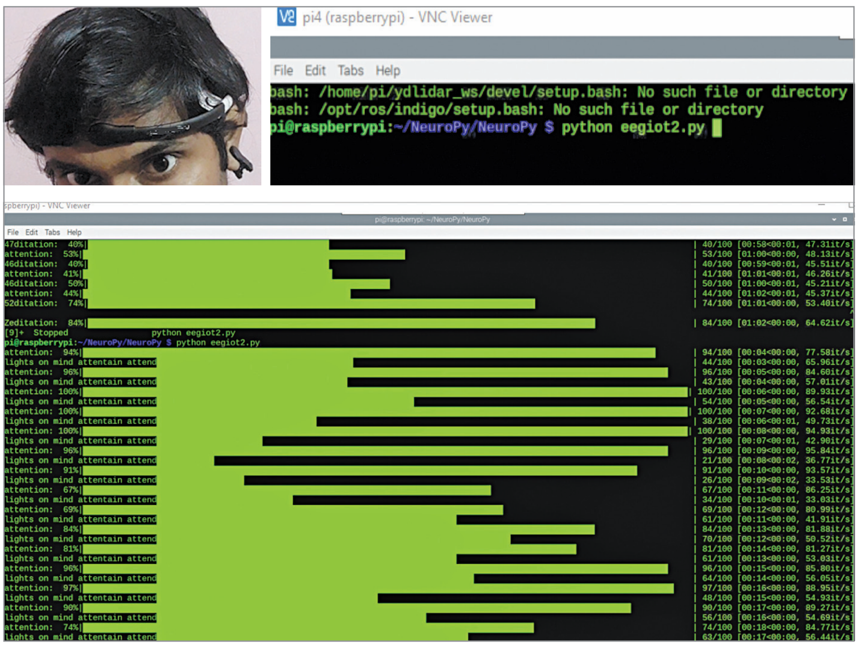

Fig. 8: Brain attention level during testing

High beta waves
High gamma waves
Low gamma waves
Attention
Meditation
Waves
Eyeblink

To control the connected IoT devices such as electric lights or fans, use the Attention, Meditation, and Eyeblink data.

Coding

Include the NeuroPy and gpiozero Python modules in the code. Then set the COM port name for the EEG sensor. Next, set the pin number for controlling the electric lights, fans, etc.

Now create a function that checks the brain signals/waves like alpha waves, beta waves, gamma waves and so on, so that your attention value can be determined. If it exceeds the pre-set threshold value of 80 while thinking of switching on/off the desired electric device (in this case electric lights), it will occur automatically—like magic.

Connection and testing

After making the appropriate connections, run the code in an IDE or terminal and concentrate on switching a light bulb on or off with your thoughts. When your thoughts reach or exceed the set threshold value, the light bulb will glow instantly.

Congrats! You have just controlled an electric device using your thoughts. EFY

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